COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	Jul 9, 2026	**DATE SUBMITTED:**	
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ACTIVITY NO. 3P1 **Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A:

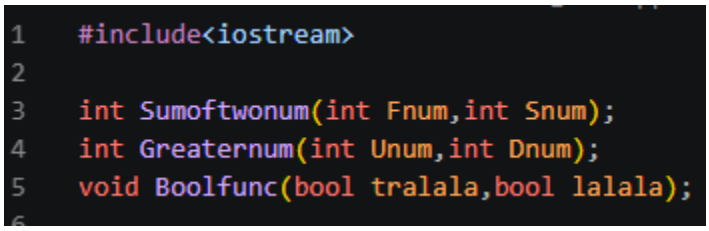
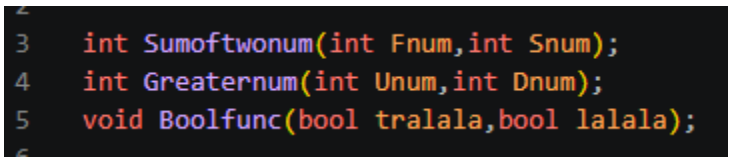
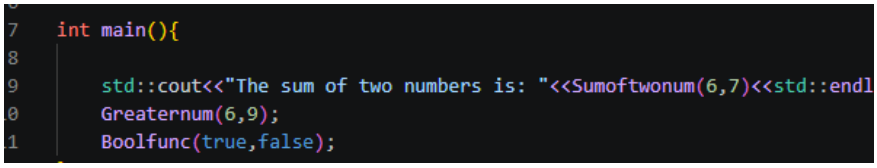
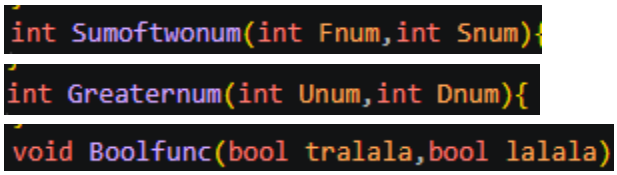
```cpp
#include<iostream>
#include<iomanip>

int Sumoftwonum(int Fnum,int Snum);
void Greaternum(int Unum,int Dnum);
void Boolfunc(bool tralala,bool lalala);

int main(){

    std::cout<<"The sum of two numbers is: "<<Sumoftwonum(6,7)<<std::endl;
    Greaternum(6,9);
    Boolfunc(true,false);
}
int Sumoftwonum(int Fnum,int Snum){
    int sum=Fnum+Snum;
    return sum;
}
void Greaternum(int Unum,int Dnum){
    if(Unum<Dnum){
        std::cout<<Unum<<" is less than "<<Dnum<<std::endl;
    }else{
        std::cout<<Unum<<" is greater than "<<Dnum<<std::endl;
    }
}
void Boolfunc(bool tralala,bool lalala){
    std::cout << "Logical AND (a && b):    " << (tralala && lalala) << std::endl;
    std::cout << "Logical OR (a || b):     " << (tralala || lalala) << std::endl;
    std::cout << "Logical NOT (!a):        " << (!tralala) << std::endl;
    std::cout << "Logical NOT (!b):        " << (!lalala) << std::endl;
    std::cout << "Logical XOR (a ^ b):     " << (tralala ^ lalala) << std::endl;
}
```

```
The sum of two numbers is: 13
6 is less than 9
Logical AND (a && b):    0
Logical OR (a || b):     1
Logical NOT (!a):        0
Logical NOT (!b):        1
Logical XOR (a ^ b):     1
```

Sections	Answer
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	N/A
Main Function	
Method Definition	

ILO B:

```cpp
#include <iostream>

class Triangle{
    private:
        double totalangle,angleA,angleB,angleC;
    public:
        Triangle(double A, double B,double C);
        void setAngles(double A,double B,double C);
        const bool validateTriangle();

};

int main(){
    Triangle set1(60,90,30);
    if(set1.validateTriangle()){
        std::cout<<"The shape is a valid triangle \n";
    }else{
        std::cout<<"The shape is NOT a valid triangle \n";
    }

    return 0;
}
Triangle::Triangle(double A,double B,double C){
    angleA=A;
    angleB=B;
    angleC=C;
    totalangle=A+B+C;
}
void Triangle::setAngles(double A,double B,double C){
    angleA=A;
    angleB=B;
    angleC=C;
    totalangle=A+B+C;
}
const bool Triangle::validateTriangle(){
    return(totalangle<=180);
}
```

```
The sum of two numbers is: 13
6 is less than 9
Logical AND (a && b):    0
Logical OR (a || b):     1
Logical NOT (!a):        0
Logical NOT (!b):        1
Logical XOR (a ^ b):     1
```

It outputs as expected from the main structure's parameters. It correctly initiates the function to validate the three values if it counts as a triangle.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO 3:

1.
```
void SN(double &val1, double &val2) {
    double temp = val1;
    val1 = val2;
    val2 = temp;
}
```

```
double firstNum = 10.5, secondNum = 25.8;
cout << "Original: A = " << firstNum << ", B = " << secondNum << endl;
SN(firstNum, secondNum);
cout << "Swapped: A = " << firstNum << ", B = " << secondNum << endl;
```

```
Original: A = 10.5, B = 25.8
Swapped: A = 25.8, B = 10.5
```

2.
```
double KTF(double kelvin) {
    return (kelvin - 273.15) * 9.0 / 5.0 + 32.0;
}
```

```
double originalKelvin = 300.0;
cout<<originalKelvin<<" Kelvin = "<<KTF(originalKelvin)<<" Fahrenheit"<<endl;
```

```
300 Kelvin = 80.33 Fahrenheit
```

3.
```
double CD(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}
```

```
double ptA_x = 0, ptA_y = 0;
double ptB_x = 3, ptB_y = 4;
double lineLength = CD(ptA_x, ptA_y, ptB_x, ptB_y);
cout << "The straight-line distance is: " << lineLength << endl;

double vertexA_x = 0, vertexA_y = 0;
double vertexB_x = 4, vertexB_y = 0;
double vertexC_x = 2, vertexC_y = 3;

double side1 = CD(vertexA_x, vertexA_y, vertexB_x, vertexB_y);
double side2 = CD(vertexB_x, vertexB_y, vertexC_x, vertexC_y);
double side3 = CD(vertexC_x, vertexC_y, vertexA_x, vertexA_y);

cout << "Calculated sides: " << side1 << ", " << side2 << ", " << side3 << endl;
cout << "Total Area      : " << CA(side1, side2, side3) << endl;
cout << "Total Perimeter : " << CP(side1, side2, side3) << endl;
```

```
The straight-line distance is: 5
Calculated sides: 4, 3.60555, 3.60555
Total Area      : 6
Total Perimeter : 11.2111
```

4.

a.

```
double Triangle::CA() {
    double s = (sideA + sideB + sideC) / 2.0;
    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}
```

```
Area        : 6
```

b.

```
double Triangle::CP() {
    return sideA + sideB + sideC;
}
```

```
Perimeter    : 12
```

c.

```
string Triangle::CT() {
    if (angleA < 90 && angleB < 90 && angleC < 90) {
        return "Acute-angled";
    } else if (angleA > 90 || angleB > 90 || angleC > 90) {
        return "Obtuse-angled";
    } else {
        return "Others (Right-angled)";
    }
}
```

```
Type        : Others (Right-angled)
```

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code explanation, debugging, and idea generation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Explained the process behind coding structures and how they work, fixing error issues during compiling, and had ideas for variables to use.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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